Mini Project Report

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COP 3635: Systems and Networks
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Project Title:

Campus Network with VLAN and inter-VLAN Routing

Objective:

Simulate a University Campus Network using multiple VLANs connected via inter-VLAN routing.

Description:

We simulated three VLANs for a typical campus network. These VLAN groups were: STUDENTS, FACULTY, and ADMIN. Each of the PCs fall under the same VLAN and share a similar IP address (such as **192.168.#.10-12**). The " # " represents which VLAN tag the PC belongs to (10 for students, 20 for faculty, and 30 for the administration). The leftmost bits depict the exact PC in that VLAN.

Communication between the different VLANs occurs using inter-VLAN routing. Once the packets travel through their switch, they are assigned a VLAN tag depending on which VLAN they originate from and where they plan on traveling. The multilayer/connector switch allows the transportation of packets across the different switches. The router serves as the "brains" of the operation, as it determines where the packets have to go and how to get them to their intended destination.

Two tests are run in order to test the functionality of the VLANs as well as their ability to communicate with one another. Test one simply involves establishing a connection between one PC and the router. This test is done in order to ensure that the packets can even reach the brain in the first place, which is essentially one half of the entire operation. The second test involves transferring packets from a PC on one VLAN to a PC on a different VLAN. This process involves the packet traveling to the router, and being sent to the different VLAN PC correctly. One thing of note is that on the first iteration of test two the packet fails to reach its intended destination. However, on the second and all other subsequent tests, the process is successful. This is due to the connection not fully being established during the first iteration. But since in the second iteration the connection is established, the packets will successfully reach the other VLANs PC every time.

OSI Layers Utilized:

- **Layer 1 (Physical)** - Cables & ports used to form the network. Crucial for trunk links that connect the individual VLAN switches to the multilayer router and switch

- **Layer 2 (Data Link)** - For the individual VLANs. VLAN IEEE tagging (802.1Q) occurs here, MAC address learning, and framing.

- **Layer 3 (Network)** - For Inter-VLAN Routing. Sends traffic to a Layer 3 device (router or switch), uses IP addressing to route packets between subnets.

Contributions:

Daniel Lowry:

- VLAN & inter-VLAN research
- Refining network topology
- Testing network functionality
- PowerPoint presentation construction
- Refining Project Report

Nicholas Palceski:

- VLAN & inter-VLAN research
- Network topology construction
- Testing network functionality
- Refining PowerPoint presentation
- Project report Construction

Configuration:

1. The first thing that needs to happen is to set up the VLANs for every switch so they recognize which VLANS exist on the network.

```
1. All Switches:

enable
configure terminal

vlan 10
name STUDENTS

vlan 20
name FACULTY

vlan30
name ADMIN

end
write memory
```

2. Next, at each department switch, we configure each specific PC cable connected to that switch to their respective VLAN (students set to 10, admin set to 30, etc).

```
2. Department Switches:

STUDENT:

enable
configure terminal
interface range fa0/2 - 4
switchport mode access
switchport access vlan 10
end
write memory

FACULTY:

enable
configure terminal
interface range fa0/2 - 4
switchport mode access
switchport access vlan 20
end
write memory

ADMIN:

enable
configure terminal
interface range fa0/2 - 4
switchport mode access
switchport access vlan 30
end
write memory
```

3. Following that, we configure the core switch (handles incoming and outgoing packets from departments and router) to enable trunk connection to the other switches as well as the router. This allows VLANS to “talk” with each other.

```
3. Core Switch:

enable
configure terminal
interface range fa0/1 - 3
switchport mode trunk
interface g0/1
switchport mode trunk
end
write memory
```

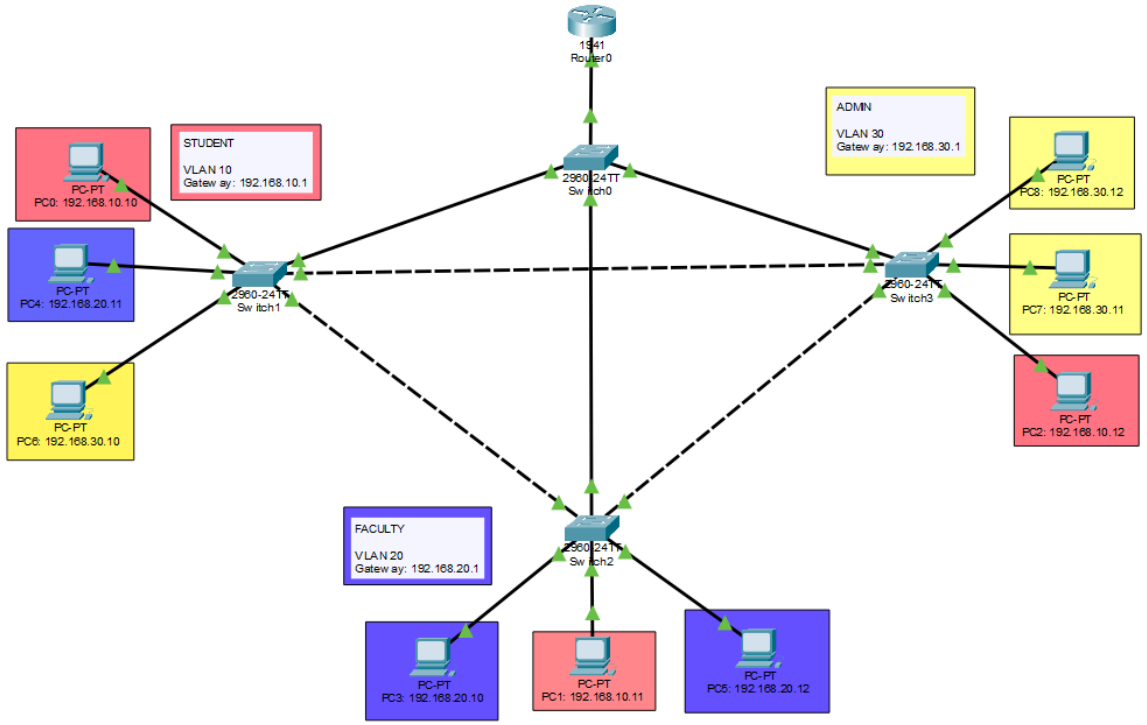
4. Same as before, we are setting up the connections to be trunk for the department switches. This involves setting the trunk for the connection line between department switch and core switch.

```
4. Department switches:

enable
configure terminal

interface fa0/1
switchport mode trunk

end
write memory
```

5. Lastly, we configure the router to keep running, then we configure it to know about the existing VLANs on the network and which connection line or address it will use to send and receive data from.

```
5. Router:

enable
configure terminal

interface g0/0
no shutdown

interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0

interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0

interface g0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0

end
write memory
```

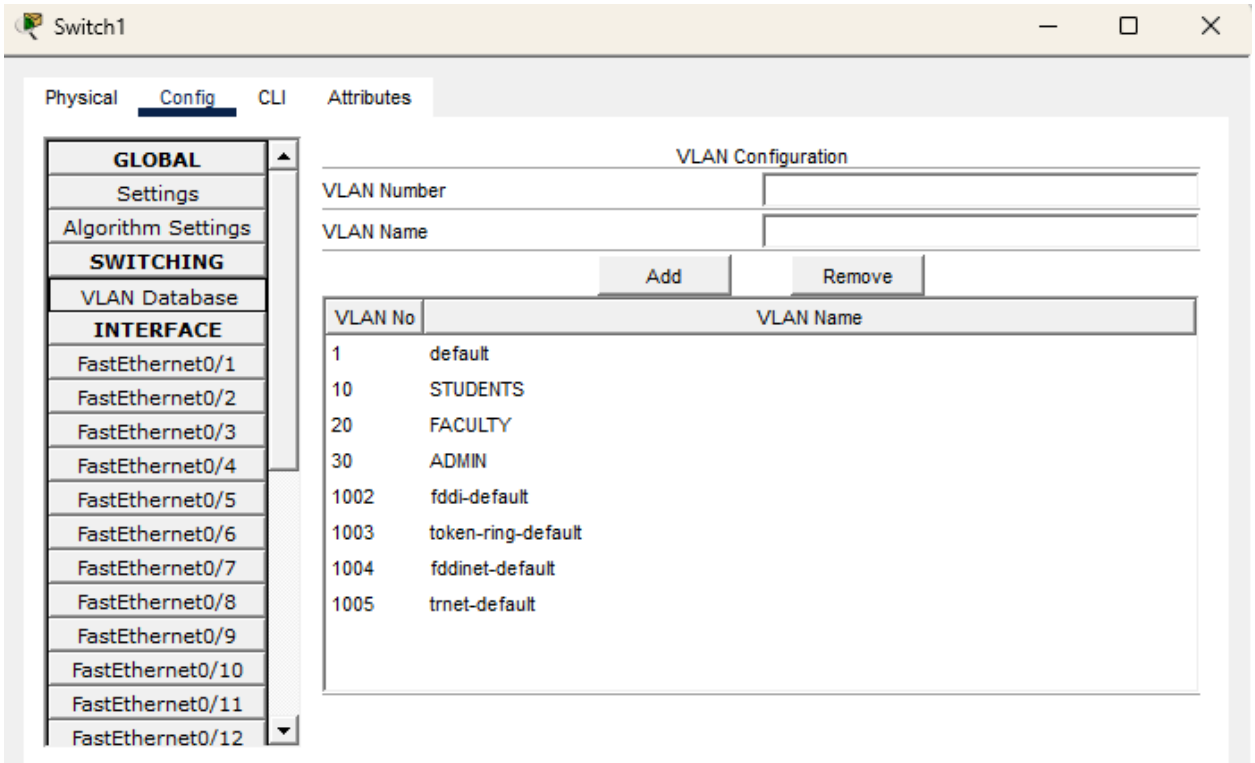
Network Topology:



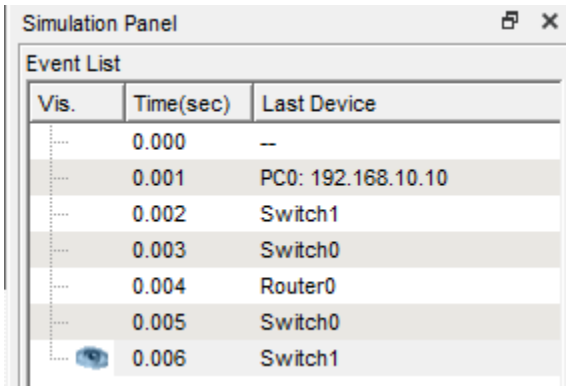
PC0 Configuration:

```
Device Name: PC0: 192.168.10.10
Device Model: PC-PT

Port            Link   IP Address         IPv6 Address                          MAC Address
FastEthernet0   Up     192.168.10.10/24   <not set>                             0040.0B77.89C7
Bluetooth       Down   <not set>          <not set>                             00D0.BCA1.846A

Gateway: 192.168.10.1
DNS Server: <not set>
Line Number:  <not set>

Physical Location: Intercity > Home City > Corporate Office > PC0: 192.168.10.10
```

VLAN Database (consistent across each switch):



Test 1 – Establishing connection between PC0 and the router:



Test 2 – Sending packets across VLANs using inter-VLAN routing (in this case sending packets from PC0 (Student) → PC3 (Faculty)):

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Failed	PC0: ...	PC3: 192.1...	ICMP		0.000	N	0	(e...	(delete)
	Successful	PC0: ...	PC3: 192.1...	ICMP		0.000	N	1	(e...	(delete)

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device
	0.001	--	PC0: 192.168.10.10
	0.002	PC0: 192.168.10.10	Switch1
	0.002	Switch1	Switch0
	0.003	Switch1	Switch0
	0.003	Switch0	Router0
	0.004	Switch0	Router0
	0.004	Router0	Switch0
	0.005	Router0	Switch0
	0.005	Switch0	Switch2
	0.006	Switch0	Switch2
	0.006	Switch2	PC3: 192.168.20.10
	0.007	Switch2	PC3: 192.168.20.10
	0.007	PC3: 192.168.20.10	Switch2
	0.008	PC3: 192.168.20.10	Switch2
	0.008	Switch2	Switch0
	0.009	Switch2	Switch0
	0.009	Switch0	Router0
	0.010	Switch0	Router0
	0.010	Router0	Switch0
	0.011	Router0	Switch0
	0.011	Switch0	Switch1
	0.012	Switch0	Switch1
	0.012	Switch1	PC0: 192.168.10.10
	0.013	Switch1	PC0: 192.168.10.10
	0.867	--	Switch3

Reset Simulation ☑ Constant Delay

Captured to: 0.867 s

Conclusion:

VLANs are an excellent tool to utilize in environments where there are computers relating to different groups next to each other. As, the benefits of a LAN can be applied without the major restriction of the systems being in close proximity to each other. Inter-VLAN routing is the other critical component of VLANs that you can essentially not live without if you are using VLANs. The two primary methods used in inter-VLAN routing: router-on-a-stick and switch virtual interfaces are great in their respective areas. ROS being a greater cost effective choice for smaller setups, and SVIs being a great choice for those interested in better performance and scalability for larger, more complex environments.

This project has done wonders for our team's own personal knowledge as well as our careers. It has allowed us to gain real, applicable knowledge about a specific aspect of computer networks. Additionally, it showed us how VLANs and inter-VLAN routing are incorporated into more complex systems and the Internet as a whole. We are grateful to have the opportunity to complete a project such as this, even under a smaller than normal amount of time. VLANs and their communication methods have intrigued our team and we plan on delving deeper into them and network communication as a whole after the course comes to an end.